

Project Design Phase-II Technology Stack (Architecture & Stack)

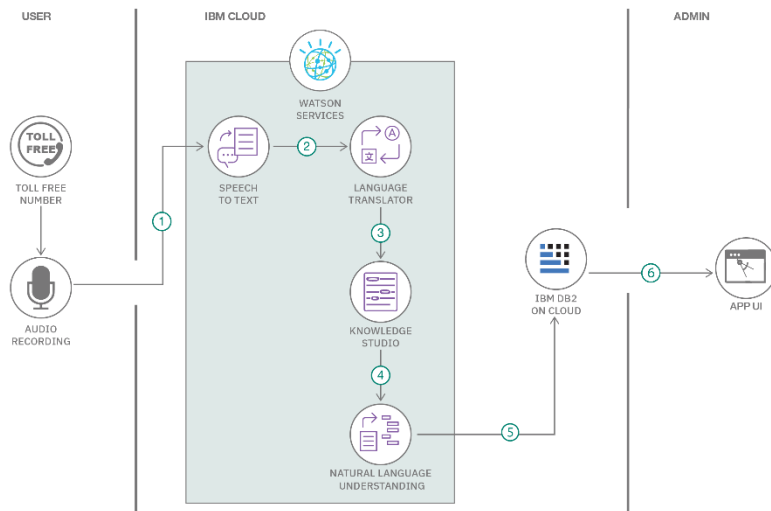
Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Single-page interface encompassing a highly fluid customer catalog storefront and a unified, dark-themed admin dashboard.	React.js (v18+), Vite, Optimized Vanilla CSS Variables.
2.	Application Logic-1	Core backend logic tier running route controllers to sanitize payloads, match text parameters, and process transactions.	Node.js, Express.js RESTful API.
3.	Application Logic-2	Dynamic layout theme modification engine that injects an .admin-theme override class directly into the document body tag.	React Router v6, Global React Context hooks.
4.	Application Logic-3	Stateless session retention layer that signs, transmits, and validates client authentication tokens.	JSON Web Tokens (JWT), Browser LocalStorage.
5.	Database	Object Data Modeling (ODM) tier incorporating a custom proxy pattern to cast raw 24-character hex strings into native system IDs.	Mongoose ORM, makeModelProxy interceptor pattern.
6.	Cloud Database	Fully managed NoSQL cloud environment housing user documents, catalog details, active carts, and order entries.	MongoDB Atlas.
7.	File Storage	Storage framework managing image path URLs for hero promotional banners, main item cards, and gallery thumbnail carousels.	Local Server File System / Cloud Object Storage.
8.	External API-1	Text regular expression engine designed to parse keyword search inputs against titles and category documents dynamically.	Native Express RegExp query parsing.
9.	External API-2	Cloud typographic distribution pipeline serving clean, high-performance web font elements to the presentation tier.	Google Fonts API (Outfit typography).
10.	Machine Learning Model	Relies on deterministic schema categories to sort and match listings.	JavaScript filtering algorithms.
11.	Infrastructure (Server / Cloud)	Process-clustered containers deploying decoupled client-server builds to support isolated cloud expansions.	Local Host, PM2 clustering / Docker containers on AWS ECS.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The core open-source modular components driving component-based frontend routing and fast asynchronous API route handling.	React.js, React Router v6, Node.js, Express.js.
2.	Security Implementations	One-way cryptographic hashing for passwords, coupled with role-aware backend middleware filters that reject unauthorized route entries with 403 blocks.	bcryptjs (10 salt rounds), Stateless JWT Bearer Authorization Headers.
3.	Scalable Architecture	Decoupled, stateless 3-Tier Architecture isolating presentation modules from routing paths, enabling independent resource horizontal scaling via containers.	MERN Stack, RESTful API endpoints, Docker, NGINX Load Balancers.
4.	Availability	Achieves high operational reliability by implementing load-balanced runtime containers paired with a local mock database file-fallback loop if Atlas timeouts trigger.	Cloud Run instance health checks, makeModelProxy database resiliency handler.
5.	Performance	Keeps initial mobile paint bundles under 250KB by using pure, framework-free CSS variable declarations, achieving backend catalog data query responses under 200ms.	Vite Asset Compression, Pure Vanilla CSS variables, image lazy loading.

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>